Applied Physics -1

Measurement Fundamentals - Lecture 01 (Module -01)

Introduction to Measurement

Definition

Measurement is the process of comparing an unknown physical quantity with a known standard quantity (unit).
Example:
Measuring the length of a table using a metre scale.

Every measurement consists of

- Numerical value
- Unit

Example:

$$Length = 5\ m$$

where
5 = Numerical value
m = Unit

Need for Measurement

Measurements are essential because they

- Ensure accuracy in engineering
- Allow comparison of quantities
- Help in manufacturing components
- Improve quality control
- Enable scientific research
- Ensure international standardization

Physical Quantities

A physical quantity is any measurable property.
Examples:

- Length
- Mass
- Time
- Temperature
- Electric Current

Units and Dimensions

Unit

A unit is a standard quantity used to measure a physical quantity.
Example:

$$Length = 10\ m$$

Here metre is the unit.

Fundamental Quantities

Definition

Fundamental quantities (also called **base quantities**) are the basic physical quantities that are **independent** of one another and cannot be expressed in terms of other physical quantities. They serve as the foundation for all other physical quantities used in science and engineering.
The International System of Units (SI) defines **seven fundamental quantities**. All other quantities, known as **derived quantities**, are obtained by combining these fundamental quantities mathematically.

Characteristics of Fundamental Quantities

- They are independent of each other.
- They cannot be derived from any other physical quantities.
- Each fundamental quantity has a unique SI unit.
- They form the basis for defining derived quantities.
- They are universally accepted through the SI system.

SI Fundamental Quantities

S.No.	Physical Quantity	SI Unit	Symbol
1	Length	metre	m
2	Mass	kilogram	kg
3	Time	second	s
4	Electric Current	ampere	A
5	Thermodynamic Temperature	kelvin	K
6	Amount of Substance	mole	mol
7	Luminous Intensity	candela	cd

Derived Quantities

Derived quantities are obtained from fundamental quantities.
Examples

$$Velocity = \frac{Length}{Time}$$

$$Acceleration = \frac{Length}{Time^2}$$

$$Force = Mass \times Acceleration$$

Dimensions

Dimensions indicate the nature of a physical quantity.
Fundamental dimensions

$$[M], [L], [T], [I], [\Theta], [N], [J]$$

where

- M = Mass
- L = Length
- T = Time
- I = Electric Current
- Θ = Temperature
- N = Amount of substance
- J = Luminous intensity

Dimensional Formula

Examples
Velocity

$$[LT^{-1}]$$

Acceleration

$$[LT^{-2}]$$

Force

$$[MLT^{-2}]$$

Work

$$[ML^2T^{-2}]$$

Pressure

$$[ML^{-1}T^{-2}]$$

Applications of Dimensional Analysis

- Verify equations
- Derive formulas
- Convert units
- Check dimensional consistency

Scientific Notation

Scientific notation expresses numbers as

$$a \times 10^n$$

where

- $1 \leq a < 10$
- n is an integer

Examples

$$4500 = 4.5 \times 10^3$$

$$0.00032 = 3.2 \times 10^{-4}$$

Advantages

- Easy calculations
- Avoids writing many zeros
- Used in engineering and science
- Reduces calculation errors

Order of Magnitude

The order of magnitude represents the nearest power of ten of a quantity.

Determination of Order of Magnitude

Any physical quantity can be expressed in scientific notation as

$$N = a \times 10^n$$

where,

- $1 \leq a < 10$ is called the **coefficient (or mantissa)**.
- n is an integer (positive, negative, or zero) called the **exponent**.

The order of magnitude is determined by the value of the coefficient a:

- If $a < 5$, then the order of magnitude is

$$\boxed{10^n}$$

- If $a \geq 5$, then the order of magnitude is

$$\boxed{10^{n+1}}$$

Example

$$3450 = 3.45 \times 10^3$$

Order of magnitude

$$10^3$$

Examples

Quantity	Order
Human height	10^0 m
Earth radius	10^6 m
Speed of light	10^8 m/s
Atom diameter	10^{-10} m

Importance

- Simplifies very large numbers
- Simplifies very small numbers
- Useful in astronomy
- Useful in nanotechnology

Significant Digits

Significant digits are the meaningful digits that indicate the precision of a measurement.
Examples

$$25.67$$

contains four significant figures.
Rules

1. All non-zero digits are significant.
2. Zeros between non-zero digits are significant.
3. Leading zeros are not significant.
4. Trailing zeros after decimal are significant.
5. Exact numbers have infinite significant figures.

Examples

Number	Significant Figures
25.40	4
0.00456	3
7000	1
7.000	4

Importance

- Indicates measurement precision
- Prevents false accuracy
- Maintains consistency

Errors in Measurement

An error is the difference between the measured value and the true value.

$$Error = True\ Value - Measured\ Value$$

Types of Errors

Gross Errors

Caused by human mistakes.
Examples

- Wrong reading
- Wrong calculation
- Recording incorrect values

Systematic Errors

Occur due to instrument or experimental setup.
Examples

- Zero error
- Calibration error
- Environmental effects

Random Errors

Occur due to unpredictable variations.
Examples

- Observer variation
- Temperature fluctuation
- Electrical noise

Accuracy and Precision

Accuracy refers to closeness to the true value.
Precision refers to closeness among repeated measurements.

Absolute Error

Absolute error is the difference between the measured value and the true value. It indicates how far the measured value deviates from the actual value.

$$\Delta x = |x_m - x_t|$$

where,

Δx Absolute error

x_m Measured value obtained from the experiment

x_t True (actual or accepted) value

$|\ |$ Absolute value, which ensures the error is always positive

Why this formula?
The error is obtained by subtracting the true value from the measured value. The absolute value is taken because the magnitude of the error is important, not whether it is positive or negative.

Relative Error

Relative error is the ratio of the absolute error to the true value. It indicates the error relative to the size of the measured quantity.

$$\text{Relative Error} = \frac{\Delta x}{x_t}$$

where,

Δx Absolute error

x_t True value

Why this formula?
Dividing the absolute error by the true value gives the error as a fraction of the actual measurement, allowing comparison of errors in quantities of different magnitudes.

Percentage Error

Percentage error expresses the relative error as a percentage.

$$\%\ \text{Error} = \frac{\Delta x}{x_t} \times 100$$

where,

Δx Absolute error

x_t True value

100 Converts the relative error into percentage

Why this formula?
Multiplying the relative error by 100 converts it into a percentage, making it easier to interpret and compare the accuracy of different measurements.

Propagation of Errors

Propagation of errors is the method of determining how the uncertainty (error) in measured quantities affects the uncertainty of the final calculated result.

Where,
Z : Calculated quantity
A, B : Measured quantities
ΔA, ΔB : Absolute errors in A and B
ΔZ : Absolute error in Z

1. Addition or Subtraction

If

$$Z = A \pm B$$

then

$$\Delta Z = \Delta A + \Delta B$$

Explanation:
When two quantities are added or subtracted, their absolute errors are added because the uncertainty from both measurements contributes to the final result.

2. Multiplication or Division

If

$$Z = A \times B$$

or

$$Z = \frac{A}{B}$$

then

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Explanation:
For multiplication and division, the percentage (or relative) errors are added because the error depends on the size of each measured quantity.

3. Power

If

$$Z = A^n$$

then

$$\frac{\Delta Z}{Z} = n\frac{\Delta A}{A}$$

where

$$n = \text{Power (index) of } A$$

Explanation:
When a quantity is raised to a power, its relative error is multiplied by that power. Hence, higher powers increase the uncertainty in the final result.
Note:

- Addition/Subtraction → Add absolute errors.
- Multiplication/Division → Add relative (percentage) errors.
- Power → Multiply the relative error by the power.

Engineering Applications

Robotics: Robots use accurate sensor measurements for navigation, obstacle detection, and precise motion control.
Aerospace Engineering: Precise measurements of altitude, velocity, and fuel ensure safe and successful flight missions.
Manufacturing Engineering: Accurate measurements help produce components within specified tolerances and improve quality.
Civil Engineering: Measurements ensure the safe design and construction of buildings, bridges, and roads.
Electronics Engineering: Electrical parameters such as voltage, current, and resistance are measured for circuit analysis and testing.
Medical Instrumentation: Medical devices rely on precise measurements for accurate diagnosis and patient monitoring.
Artificial Intelligence and Smart Sensors: AI systems use accurate sensor data to improve prediction, automation, and intelligent decision-making.

Key Takeaways

- Every measurement consists of a numerical value and a unit.
- SI Units ensure worldwide standardization.
- Dimensions verify physical equations.
- Scientific notation simplifies calculations.
- Significant figures indicate precision.
- Errors are unavoidable but can be minimized.
- Error propagation helps estimate uncertainty in calculations.

Worksheet Questions

1. Explain the importance of measurement in engineering. Discuss why standard units are essential for scientific and engineering applications.

2. Differentiate between fundamental quantities and derived quantities with suitable examples. Write the SI units of any five fundamental quantities.

3. Determine the dimensional formula of the following quantities:
   (a) Velocity
   (b) Acceleration
   (c) Force
   (d) Work
   (e) Pressure

4. Express the following quantities in scientific notation and determine their order of magnitude:
   (a) 0.000456
   (b) 7,850,000
   (c) 299,792,458
   (d) 0.000000082

5. A metal rod has a measured length of 25.60 cm, while its true length is 25.50 cm.
   Calculate:
   (a) Absolute Error
   (b) Relative Error
   (c) Percentage Error

6. The radius of a circular disc is measured as

$$r = (10.0 \pm 0.2)\ cm$$

Find the percentage error in the calculated area of the disc.

7. The measured values are

$$A = (50.0 \pm 0.5), \qquad B = (20.0 \pm 0.2)$$

Calculate the maximum absolute error and relative error in

$$Z = \frac{A}{B}$$

8. Verify the dimensional correctness of the equation

$$s = ut + \frac{1}{2}at^2$$

using dimensional analysis.

9. During a space mission, one engineering team uses SI units while another team uses Imperial units. Explain how improper unit conversion could lead to mission failure. Suggest engineering practices that can prevent such errors.

10. A cube has side

$$(5.00 \pm 0.02)\ cm$$

Determine:

(a) Percentage error in the volume

(b) Maximum absolute error in the volume

11. The loss of NASA's Mars Climate Orbiter in 1999 is considered one of the most famous engineering failures caused by measurement errors.

Research and briefly explain:

- What actually happened?
- Which units were incorrectly used?
- What lessons should engineers learn from this incident?